Question	Answer	Marks	Partial Marks
2(b)(iii)	y = (4.4 to 5.6)x – (1.8 to 2.2) or [y =] *their* **(b)(ii)**x + *their*(y-intercept)	**2**	**FT** for any line but not horizontal or vertical line for 2 marks or **B1** **B1FT** for [m =] *their* 5 or for *their* y-intercept
3(a)	187	**2**	**M1** for $220 \times \left(1 - \frac{15}{100}\right)$ oe or **B1** for 33 seen
3(b)	19.8	**3**	**M2** for $29.7 \times \sqrt[3]{\frac{0.4}{1.35}}$ oe or **M1** for $\sqrt[3]{\frac{0.4}{1.35}}$ or $\sqrt[3]{\frac{1.35}{0.4}}$ oe seen or for $\frac{29.7^3}{x^3} = \frac{1.35}{0.4}$ oe
3(c)	12.4 or 12.44…	**3**	**M1** for $90 \times 75 \times h = 7 \times$ figs 12 **B1** for 1000 cm^3 = 1 litre soi
4(a)	32.9 or 32.91 to 32.92…	**2**	**M1** for $\pi \times 1.65 \times 4.7 + \pi \times 1.65^2$
4(b)	69.4 or 69.44 to 69.45	**2**	**M1** for cos = 1.65 ÷ 4.7 oe
4(c)(i)	12.5 or 12.54 to 12.55	**4**	**M3** for $\frac{1}{3} \times \pi \times 1.65^2 \times \sqrt{4.7^2 - 1.65^2}$ oe or **M2** for $\sqrt{4.7^2 - 1.65^2}$ oe or for $4.7 \times \sin($*their* **(b)**$)$ oe or **M1** for $1.65^2 + h^2 = 4.7^2$ oe or for $\frac{h}{4.7} = \sin($*their* **(b)**$)$ oe
4(c)(ii)	41 nfww	**4**	**B3** for 41.7… to 41.9 or **M2** for $\frac{4}{3} \times \pi \times 5^3 \div$ *their* 12.5 or **M1** for $\frac{4}{3} \times \pi \times 5^3$ After **M2** scored, **M1** for truncating *their* decimal number of cones seen to an integer answer
5(a)	$\frac{10x}{(x-3)(x+2)}$ or $\frac{10x}{x^2 - x - 6}$ final answer	**4**	**M1** for common denominator $(x-3)(x+2)$ isw **M1** for $(x+3)(x+2) - (x-2)(x-3)$ isw **B1** for correct numerator in terms of x only

Question	Answer	Marks	Partial Marks
5(b)	14	**2**	**M1** for $12-\frac{k}{2}=5$ or $2^{\frac{k}{2}}=\frac{2^{12}}{2^5}$ oe or $\frac{4096}{32}$ or $12-5$ or $2^{12}\div 2^{\frac{14}{2}}$ [= 32] seen
5(c)	$2y^3-3y^2-23y+12$ final answer	**3**	**B2** for correct unsimplified expanded expression or for simplified four-term expression of correct form with 3 terms correct or **B1** for correct expansion of 2 of the brackets with at least 3 terms correct
5(d)	$[x=]\frac{3}{y-1}$ final answer	**3**	**M1** for $xy = 3 + x$ **M1** for $xy - x = 3$ or $x-\frac{x}{y}=\frac{3}{y}$ **M1** for factorising and dividing
6(a)(i)	$\frac{1}{3}$ oe	**1**	
6(a)(ii)	100	**1**	**FT** *their* **(a)(i)** × 300 to at least 3 sf or rounded to the nearest integer
6(b)(i)	$\frac{2}{15}$ oe	**3**	**M2** for $4\times\frac{1}{6}\times\frac{1}{5}$ oe or **M1** for $k\left(\frac{1}{6}\times\frac{1}{5}\right)$ oe or list or indication of 4 correct pairs
6(b)(ii)	$\frac{3}{5}$ oe	**3**	**M2** for $1-\frac{4}{6}\times\frac{3}{5}$ or $2\left(\frac{2}{6}\times\frac{4}{5}\right)+\frac{2}{6}\times\frac{1}{5}$ oe or $\frac{2}{6}+\left(\frac{4}{6}\times\frac{2}{5}\right)$ oe or **M1** for $\frac{4}{6}\times\frac{3}{5}$ oe seen or $\frac{2}{6}\times\frac{4}{5}[\times 2]$ oe seen or $\frac{2}{6}\times\frac{1}{5}$ oe seen or correct identification of 18 pairs or space diagram oe
7(a)	$n - 5 + 3n + 10 > 105$ or better	**B1**	
	$n > 25$ final answer	**B2**	**M1** for $4n > 100$